hardware engineers) will claim the reverse: assembly is five times faster, but only 50% more difficult to use. As in most controversies, both sides can provide selective data to support their claims.

As a rule-of-thumb, expect that a subroutine written in assembly will be between 1.5 and 3.0 times faster than the comparable high-level program. The only way to know the exact value is to write the code and conduct speed tests. Since personal computers are increasing in speed about 40% every year, writing a routine in assembly is equivalent to about a two year jump in hardware technology.

Most professional programmers are rather offended at the idea of using assembly, and gag if you suggest BASIC. Their rational is quite simple: assembly and BASIC discourage the use of good software practices. Good code should be *portable* (able to move from one type of computer to another), *modular* (broken into a well defined subroutine structure), and *easy to understand* (lots of comments and descriptive variable names). The weak structure of assembly and BASIC makes it difficult to achieve these standards. This is compounded by the fact that the people who are attracted to assembly and BASIC often have little formal training in proper software structure and documentation.

Assembly lovers respond to this attack with a zinger of their own. Suppose you write a program in C, and your competitor writes the same program in assembly. The end user's first impression will be that your program is junk because it is twice as slow. No one would suggest that you write large programs in assembly, only those portions of the program that need rapid execution. For example, many functions in DSP software libraries are written in assembly, and then accessed from larger programs written in C. Even the staunchest software purist will *use* assembly code, as long as they don't have to *write* it.

Execution Speed: Hardware

Computing power is increasing so rapidly, any *book* on the subject will be obsolete before it is published. It's an author's nightmare! The original IBM PC was introduced in 1981, based around the 8088 microprocessor with a 4.77 MHz clock and an 8 bit data bus. This was followed by a new generation of personal computers being introduced every 3-4 years: 8088 → 80286 → 80386 → 80486 → 80586 (Pentium). Each of these new systems boosted the computing speed by a factor of about *five* over the previous technology. By 1996, the clock speed had increased to 200 MHz, and the data bus to 32 bits. With other improvements, this resulted in an increase in computing power of nearly *one thousand* in only 15 years! You should expect *another* factor of one thousand in the *next* 15 years.

The only way to obtain up-to-date information in this rapidly changing field is directly from the manufacturers: advertisements, specification sheets, price lists, etc. Forget books for performance data, look in magazines and your daily